

A student team building an uncrewed autonomous boat from the ground up.



Ground up means simulation + modeling first

We are building the model from first principles, mixing empirical and rule-of-thumb decisions with dynamic programming. Each constraint cuts the space of possible boats, and the optimizer finds the fastest one based on current variables. As we build and test, we keep updating the model and adding constraints: a hybrid of simulation and empirical data.

Engineers from across UW–Madison

Lead members have interned at Tesla, Northrop Grumman, Xcel Energy, Fincantieri and Milwaukee Tool.

Industrial

optimization, composites and manufacturing

Mechanical

structures, CAD

Electrical

power, battery and high voltage

Computer

embedded, firmware and software

Our primary phases

Hull

Plane fast and stay light.

Hybrid planing composite hull
Dimensions shaped by weight
Optimization shaped by hydrodynamics

Electric

Every watt-hour to speed.

High-voltage battery storage
Propulsion sized by the model
Propulsion and cooling co-designed

Autonomous

Manage drift and controls.

GPS, IMU and LIDAR sensing
Fundamental feedback protocols
Software plans the route

Timeline

Oct 25	Hull design frozen	Jan 8–30	Manufacture and assemble everything
Nov 1	Preliminary design review	Jan 8–30	Build likely spare parts, in parallel
Nov 10	Critical and long-lead parts are ordered	Jan 31	First nautical mile
Nov 20	Mechanical and electrical design frozen	Feb 1–14	Slow drives on the water
Nov 20–Dec 20	Build the hull, composite parts and any custom-milled parts	Feb 15	Start increasing speed
Jan 8	First float and tow test	Until April	Testing / Tuning

Want to get involved?

We are looking for funds, parts, advisors and supporters.
Email Marley or head to our site for more information.

bit.ly/haav

Marley Barrett, Project lead
mhbarrett@wisc.edu